



WEATHERLAB

IOT ENVIRONMENTAL MONITORING SYSTEM

Smart Weather Station & Quality Control

Continuous atmospheric sensing, local web server visualization, and live forecasting system.

PROJECT TEAM MEMBERS

David Sageder

Andrea Roithmeier

Fabian Seelig

Julian Weißböck

Matthias Winklehner

Project Assistance

David Steinegger

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Equipment

Hardware setup & time tracking.



Project Team & Roles

Clear task distribution among the 5 team members



Andrea Roithmeier

Project Lead



David Sageder

Vice Project Lead & Hardware Engineer

Raspberry Pi setup.
OpenWeather API and data logging



Fabian Seelig

API & Backend Engineer

OpenWeather API and data logging.



Julian Weißböck

Documentation

Time tracking and weekly reports.



Matthias Winklehner

Web Developer

Project webpage and admin docs.

What is Our Vision?

Compact environmental monitoring with local touch controls

Primary Goal

Build a self-contained IoT weather station providing live indoor air quality readings and outdoor forecast projections.

- ✓ **BME680 Sensor Integration**
- ✓ **Local Touch-Display GUI**
- ✓ **OpenWeather API 3-Day Forecast**

Live Environmental Data

Continuous tracking of room temperature, humidity, pressure, and air quality.

Integrated Web Server

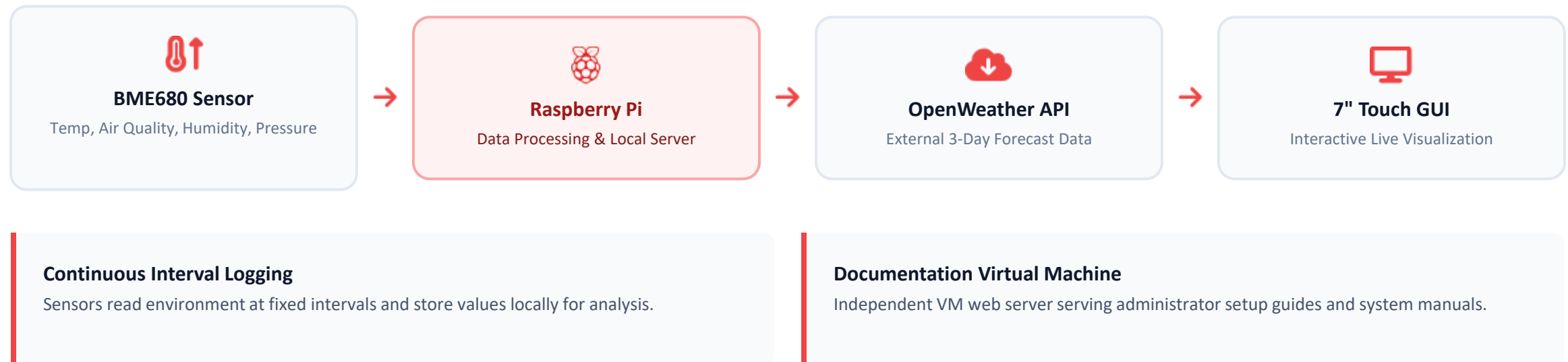
Raspberry Pi hosts local Web-GUI accessible directly on the 7-inch touch screen.

Documentation Web Portal

Dedicated Virtual Machine hosting admin setup guides and smart home features.

System Architecture & Flow

Data pipeline from hardware sensors to touchscreen presentation



Smart Home & Future Outlook

Potential expansions and advanced automation integrations



Telegram Bot Warnings

Automated instant notifications for severe weather alerts, storm warnings, or sudden temperature drops.

CAN Feature



Air Quality LED Feedback

Physical LED indicator changing color based on BME680 indoor air quality index readings.

CAN Feature



Smart Home Hub

Centralized control platform for additional home automation devices and remote web access.

Smart Home Integration

Technical Requirements

Breakdown of mandatory core features and optional enhancements

MUST Requirements

MANDATORY

- ✓ **Sensor Hardware Hookup:**
BME680 wired to Raspberry Pi for continuous reading.
- ✓ **Local Web-GUI & Display:**
Live presentation on 7-inch touch screen via local server.
- ✓ **External API Integration:**
Fetch 3-day weather forecast using OpenWeather API.
- ✓ **VM Project Portal:**
Documentation, hardware introduction, and admin guides.

CAN Requirements

OPTIONAL

- + **Telegram Alerts:**
Automated storm and extreme temperature warnings.
- + **Smartphone Webview**
A Webview of the Forecast Interface for Smartphones
- + **Visual LED Indicator:**
Physical air quality light feedback signal.

Cost

Breakdown of mandatory core features and optional enhancements

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Hardware & Equipment Specs

Physical components and hardware configuration



Raspberry Pi Unit

Provided by David Sageder

Main computing controller running local web server and database logging scripts.



BME 680 Sensor

4-in-1 Environmental Sensor

Measures temperature, humidity, atmospheric pressure, and indoor air quality.



7-Inch Touch Display

Interactive Interface

Direct touch display mounting on Raspberry Pi for real-time station monitoring.



256 GB SD Storage

High Capacity Memory

Hosts Linux OS, web server files, logs, and historical weather data archives.

Project Management Standards

Structured execution, quality assurance, and weekly tracking



Organigram

Formal team structure and clear role descriptions for each member.

MANDATORY



CPM Network Plan

ProjectLibre plan identifying critical paths and milestones.

MANDATORY



Weekly Protocols

Signed weekly activity reports documenting progress and targets.

MANDATORY



Time Tracking

Kimai software logging personal workload and hourly effort.

MANDATORY

Project Roadmap & Milestones

Phased progression from initial setup to final presentation

COMPLETE TASK LIST (25 ITEMS)

ID	Task Name	Duration	Start Date	Finish Date
	Role Allocation	1 day	15.09.26 08:00	15.09.26 17:00
	Project Topic Meeting	1 day	22.09.26 08:00	22.09.26 17:00
	Raspberry Pi	8 days	22.09.26 08:00	10.11.26 17:00
	↳ OS Installation	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Apache Web Server	2 days	29.09.26 08:00	06.10.26 17:00
	↳ Webpage Configuration	1 day	13.10.26 08:00	13.10.26 17:00
	↳ Wi-Fi Setup	1 day	20.10.26 08:00	20.10.26 17:00
	↳ Connect Display & Sensors	1 day	20.10.26 17:00	27.10.26 17:00
	↳ Configure Display & Sensors	2 days	03.11.26 08:00	10.11.26 17:00
	Documentation	9 days	15.09.26 08:00	10.11.26 17:00
	↳ Activity Logging	9 days	15.09.26 08:00	10.11.26 17:00
	↳ Technical Documentation	9 days	15.09.26 08:00	10.11.26 17:00
	Web Server	3 days	22.09.26 08:00	06.10.26 17:00
	↳ Set up Linux VM	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Network Configuration	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Apache Server Installation	1 day	22.09.26 08:00	22.09.26 17:00
	↳ Website Configuration	3 days	22.09.26 08:00	06.10.26 17:00
	Organizational Matters	9 days	15.09.26 08:00	10.11.26 17:00
	↳ Project Charter / Spec Sheet	1 day	15.09.26 08:00	15.09.26 17:00
	↳ Intro Presentation	1 day	29.09.26 08:00	29.09.26 17:00
	↳ Detailed Requirement Spec	6 days	22.09.26 08:00	27.10.26 17:00
	↳ Final Presentation Prep	8 days	22.09.26 08:00	10.11.26 17:00
	↳ Written Invitation	1 day	03.11.26 09:00	10.11.26 09:00
	↳ Topic Presentation	1 day	29.09.26 08:00	29.09.26 17:00
	↳ Final Presentation Event	0.87 days	10.11.26 09:00	10.11.26 17:00

WeatherLab – Project Timeline

Complete overview of all 25 project tasks

TOTAL TASKS 25 Tasks	MAIN GROUPS 5 Packages	PROJECT START Sep 15, 2026	PROJECT FINISH Nov 10, 2026
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VISUALIZED PROJECT SCHEDULE (SEP – NOV 2026)

ID & Task Name	September 2026	October 2026	November 2026
Role Allocation	Sep 15		
Project Topic Meeting	Sep 22		
Raspberry Pi	Sep 22 – Nov 10		
OS Installation	Sep 22		
Apache Web Server	Sep 29 – Oct 06		
Webpage Configuration		Oct 13	
Wi-Fi Setup		Oct 20	
Connect Display & Sensors		Oct 20 – Oct 27	
Configure Display & Sensors			Nov 03 – Nov 10
Documentation	Sep 15 – Nov 10		
Activity Logging	Sep 15 – Nov 10		
Technical Documentation	Sep 15 – Nov 10		
Web Server	Sep 22 – Oct 06		
Website Configuration	Sep 22 – Oct 06		
Organizational Matters	Sep 15 – Nov 10		
Requirement Specification	Sep 22 – Oct 27		
Final Presentation Prep	Sep 22 – Nov 10		
Written Invitation			Nov 03 – Nov 10
Final Presentation Event			Nov 10



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Thank You for Your Attention!

Questions & Discussion
